WORKSHEET 7

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totalPalinWords.js

```
function totalPalinWords(text) {
    let words = text.toLowerCase().match(/\b[a-z]+\b/g) || [];
    let count = 0;

    for (let word of words) {
        let reverse = word.split("").reverse().join("");

        if (word === reverse) {
            count++;
        }
    }
    return count;
}
module.exports = totalPalinWords;
```

moreVowels.js

```
function moreVowels(text) {
    let words = text.match(/\b[a-zA-Z]+\b/g) || [];
```

```
    return words.filter(word => {
        let vowels = word.match(/[aeiou]/gi) || [];
        return vowels.length >= 3;
    });
}

module.exports = moreVowels;
```

primeNumbers.js

```
function isPrime(n) {
    if (n < 2) return false;

    for (let i = 2; i <= Math.sqrt(n); i++) {
        if (n % i === 0) return false;
    }

    return true;
}

function primeNumbers(numbers) {
    return numbers.filter(isPrime);
}

module.exports = primeNumbers;
```

main.js

```
const totalPalinWords = require("./totalPalinWords");
const moreVowels = require("./moreVowels");
const primeNumbers = require("./primeNumbers");

console.log(totalPalinWords("madam is a level word"));
console.log(moreVowels("education apple beautiful sky"));
console.log(primeNumbers([2, 4, 5, 7, 9, 11, 15]));
```

```
PS E:\MCA\WEB TECHNOLOGIES\WORK SHEET 7> node main.js
3
[ 'education', 'beautiful' ]
[ 2, 5, 7, 11 ]
```

Main.js

```
const fs = require("fs");
const totalPalinWords = require("./totalPalinWords");

fs.writeFileSync("myText.txt", "Madam and mom saw a level radar.");

let text = fs.readFileSync("myText.txt", "utf8");

console.log("Total palindrome words:", totalPalinWords(text));
```

myText.txt

Madam and mom saw a level radar.

```
PS E:\MCA\WEB TECHNOLOGIES\WORK SHEET 7\PROBLEM 2> node main.js
Total palindrome words: 5
```

analysis.js

```
function analysis(text) {
    let numbers = text.match(/(?<![\w@])\d+(?!\w)/g) || [];
    let frequency = {};

    for (let num of numbers) {
        frequency[num] = (frequency[num] || 0) + 1;
    }

    return frequency;
}
module.exports = analysis;
```

main1.js

```
const fs = require("fs");
const analysis = require("./analysis");

let text = fs.readFileSync("myAnalysis.txt", "utf8");
let result = analysis(text);

let numbers = Object.keys(result).map(Number);
```

let count = numbers.length;

console.log("Total number of plain number words are:", count, numbers);

myAnalysis.txt

The subscriber ee123@g.com has 1000 renewals in the 8th month but he has only 99 renewals in the 9th month.

